

Lesson 12 — Red Light, Green Light

Matches LEGO Teacher Portal lesson: "Safe City".

Objective (I can...): I can use a color event like a traffic signal to start and stop my machine. **CS concept:** A color reading as an event (traffic-signal style events) · **Builds on:** Lesson 11 (events as start signals) and Unit 2 (color sensor).

AI Connection (2 min, read aloud): A self-driving car treats a green light as an event that means "go" and a red light as "stop" — a color starts an action. *Ask the group:* "Why is it so important the car reacts to the right color?"

Materials: LEGO CS & AI kit + device per pair, red/green/yellow color cards. **Model reuse:** Events model from Lesson 11 — code only.

60-minute flow - Warm-up (10): "Traffic game" — green card, kids tiptoe forward; red card, freeze; yellow, wiggle in place. Fast changes for giggles. Each color is an event. - **Concept + demo (8):** Demo a color event: when the sensor sees green, the motor starts (go); when it sees red, it stops. The color is the trigger. - **Build/Code (25):** Pairs code color events — green seen → go, red seen → stop — so the machine behaves like a tiny traffic signal. *Wiggle break + swap at ~12 min.* - **Challenge (7):** Add a yellow event that makes the machine slow down or beep as a warning. - **Share + tidy-up (10):** Pairs run their traffic machine with the cards and name each color event. Model stays built.

Pair roles: Driver codes the color events, Navigator works the cards and calls the color. Swap at the wiggle break. *(Trio rotates code/cards/watch.)*

Differentiation: *Support* — just green-go and red-stop. *Extension / fast-finisher* — add yellow as a third event. *Reteach* — set up green-go first, test, then add red-stop.

Exit check: Pair's machine starts and stops correctly from color events. **At-home (optional):** Watch a crossing signal; tell a grown-up what each color tells people to do. ~5 min.
